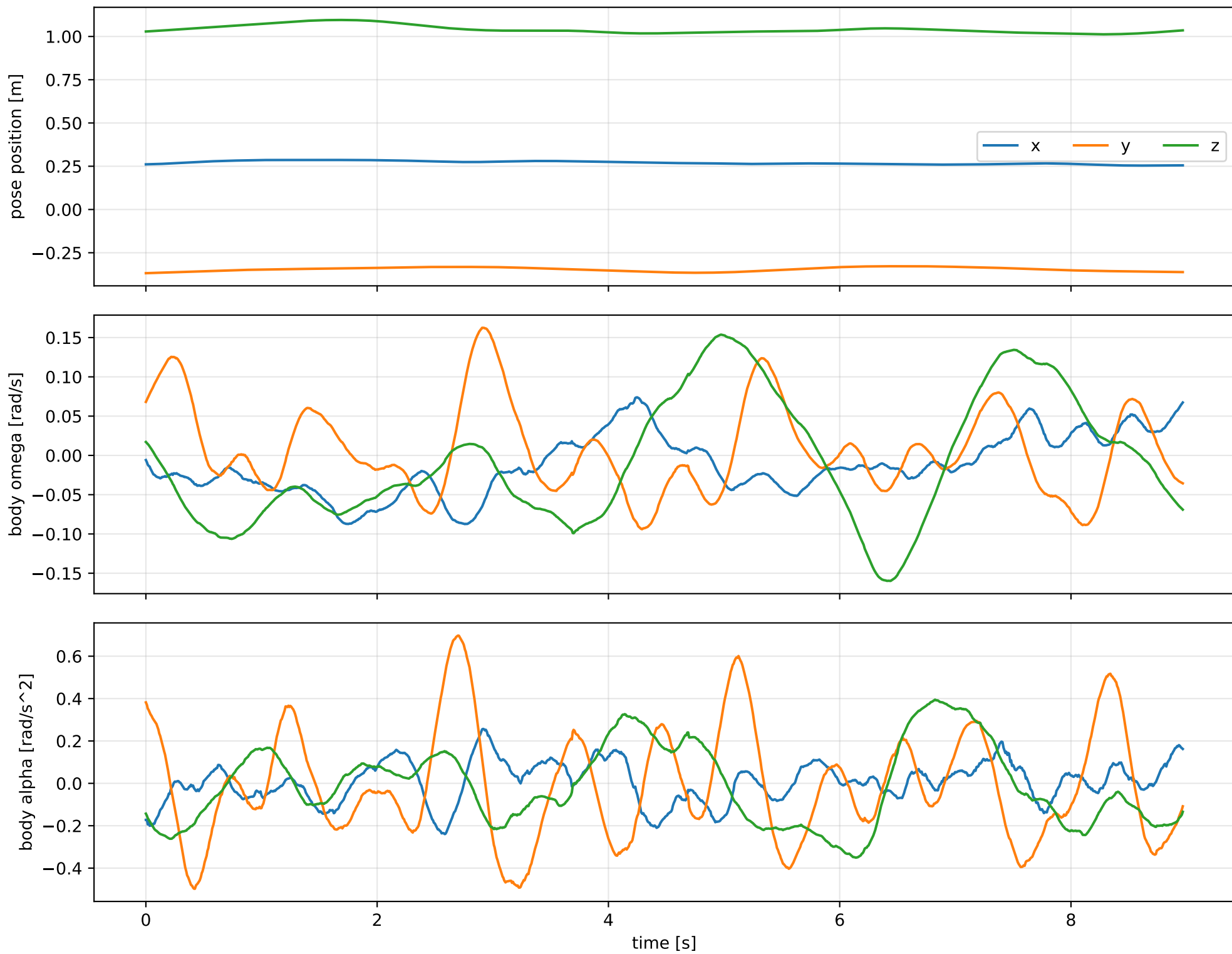
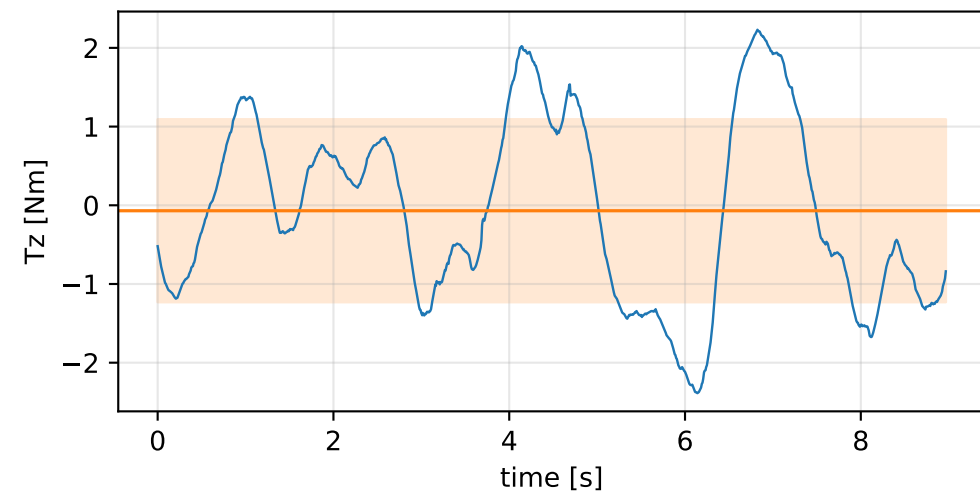
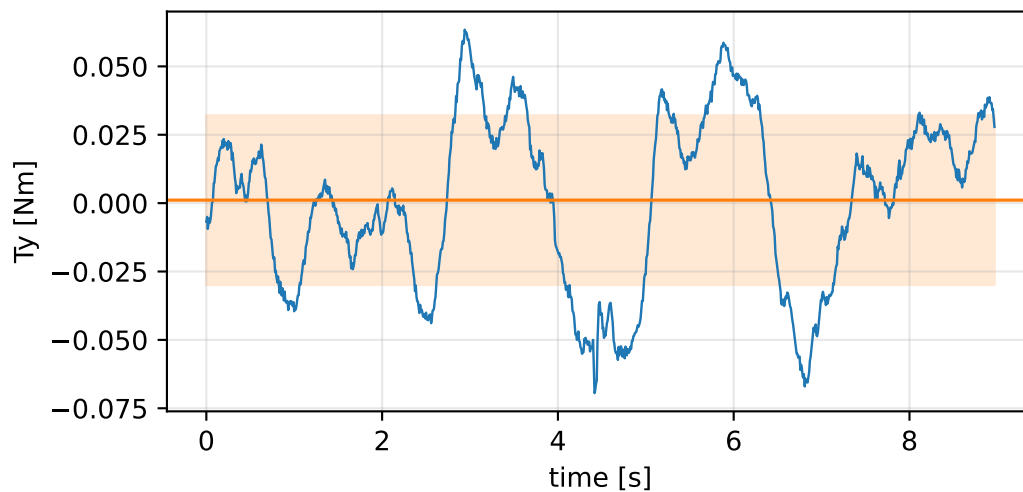
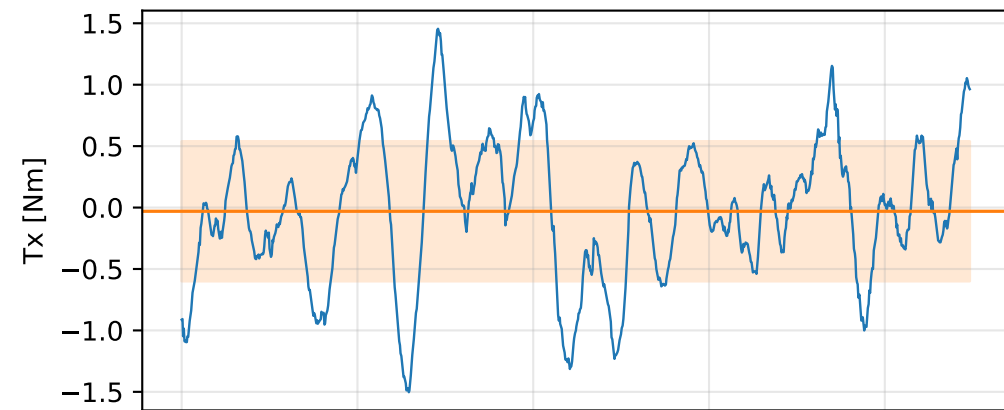
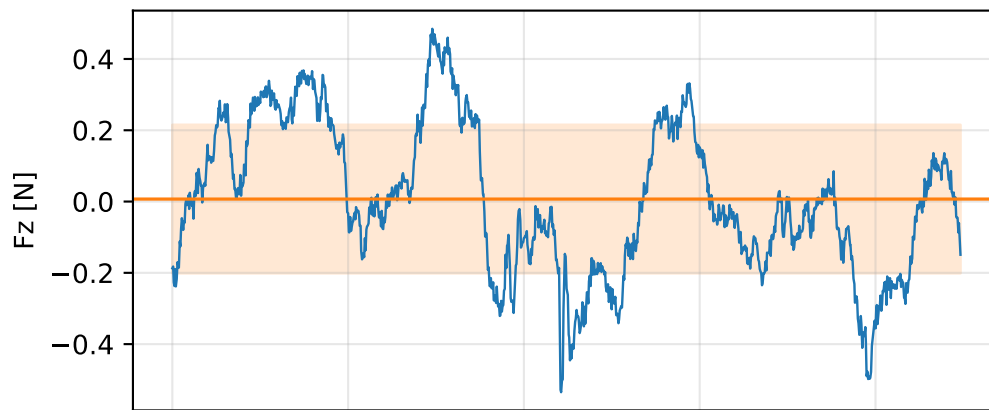
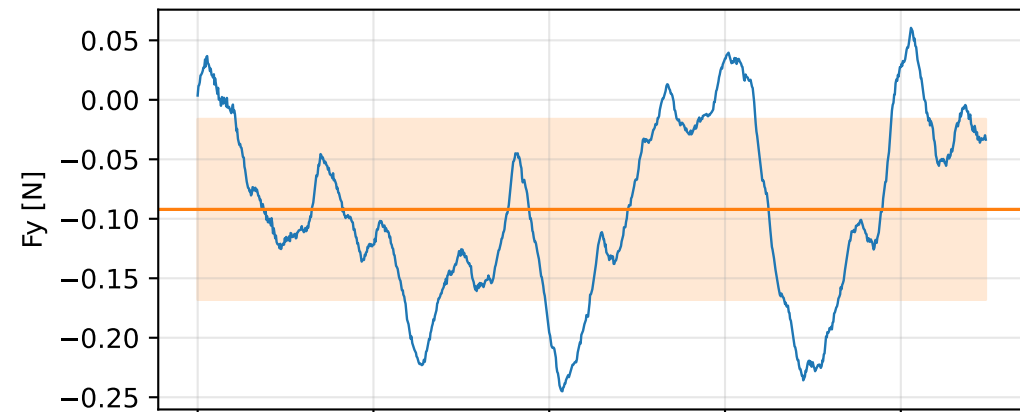
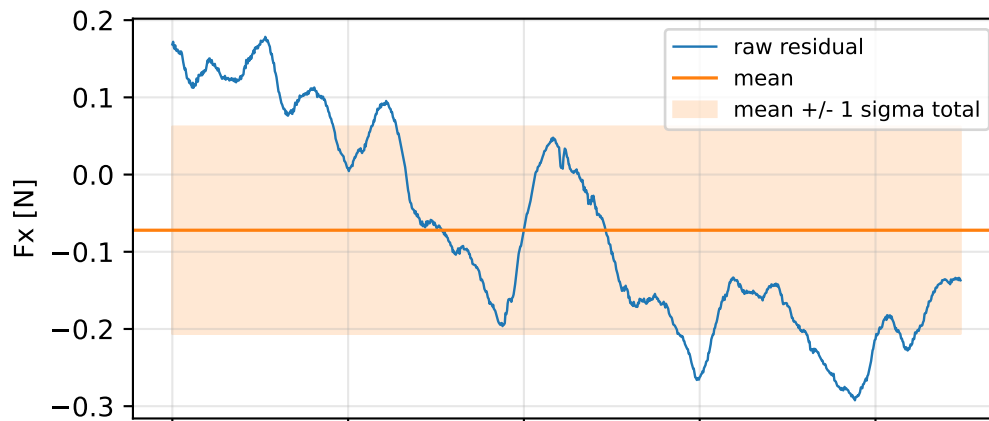


inertia_over_mass_yy_nominal: SG trajectory and derived motion



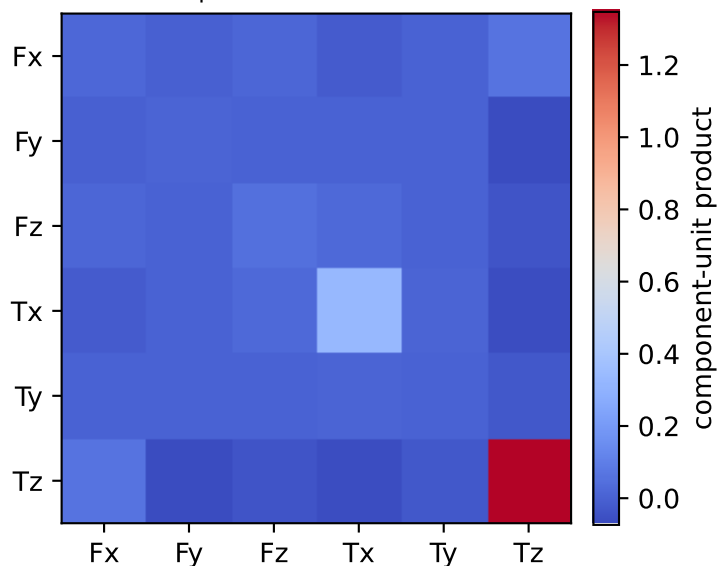
inertia_over_mass_yy_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



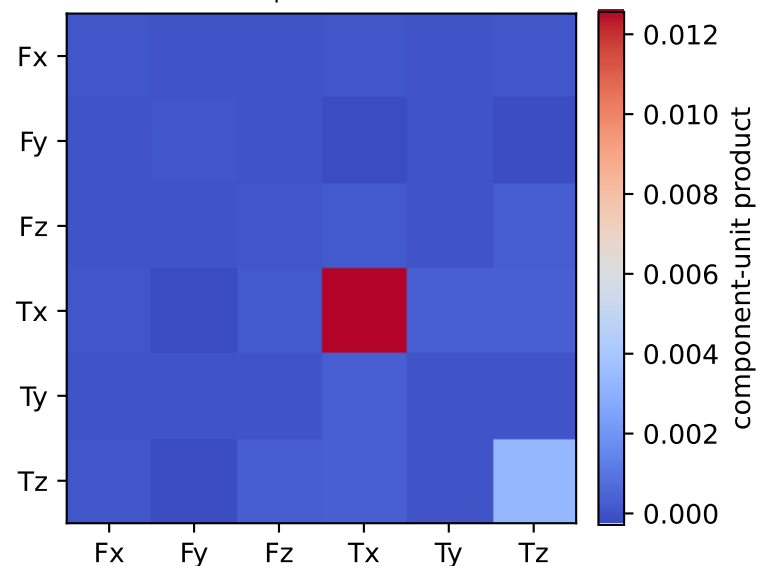
vector RMS about zero: force 0.28414 N, torque 1.2946 Nm; component std about mean: force [0.134 0.0759 0.2082], torque [0.569 0.0309 1.1607]

inertia_over_mass_yy_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [4.534e-05 1.186e-03 9.944e-03 4.556e-02 3.093e-01 1.357e+00]

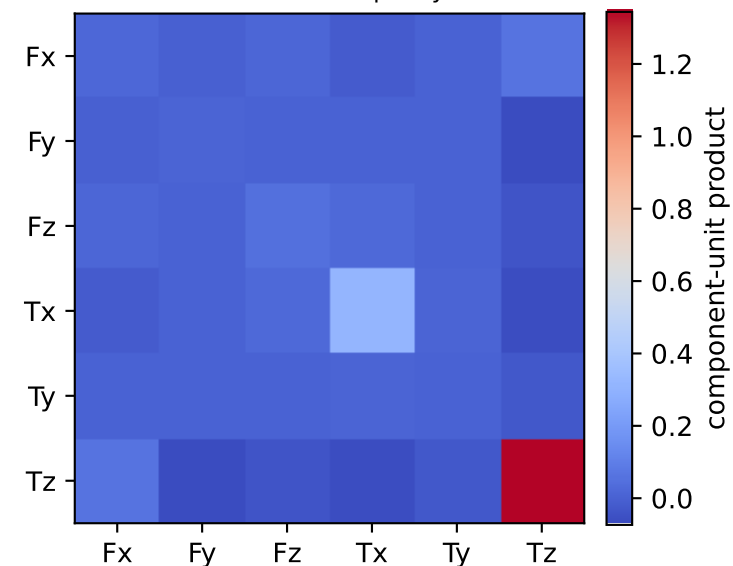
empirical total covariance



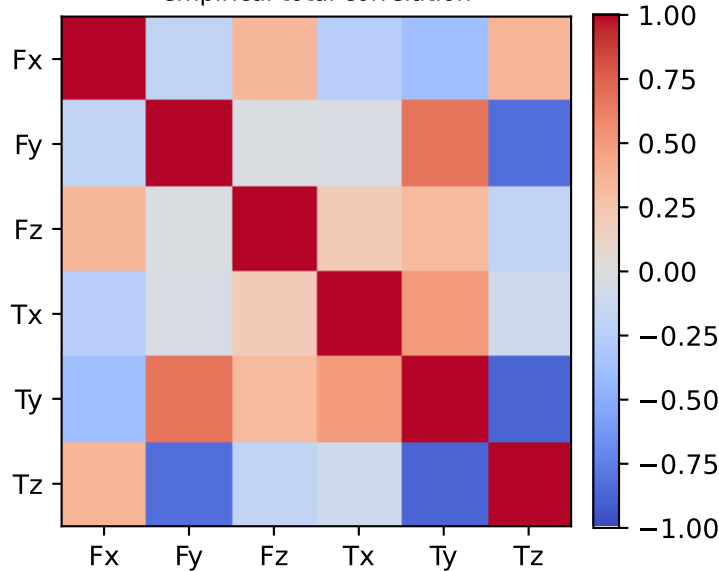
mean full-SG prediction covariance



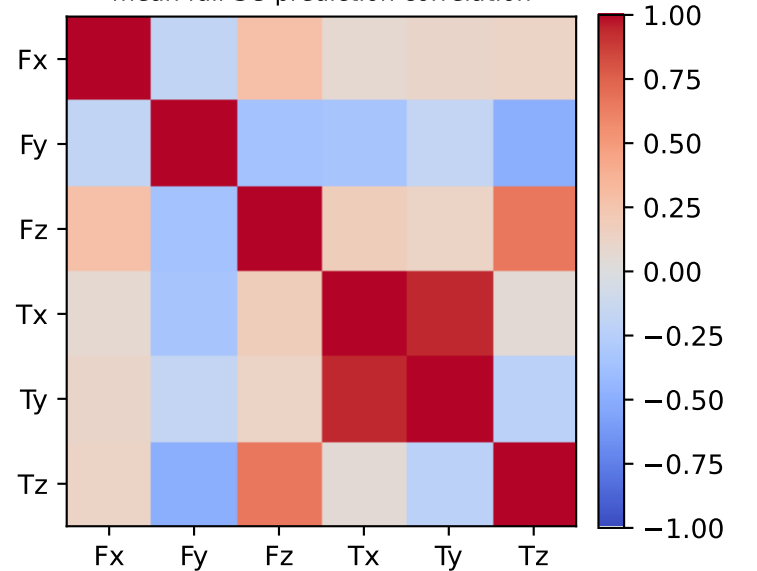
PSD excess model discrepancy covariance



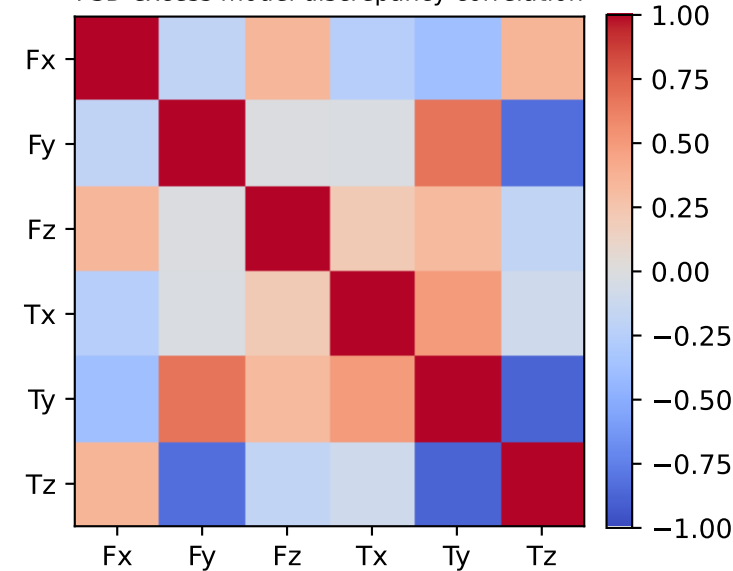
empirical total correlation



mean full-SG prediction correlation

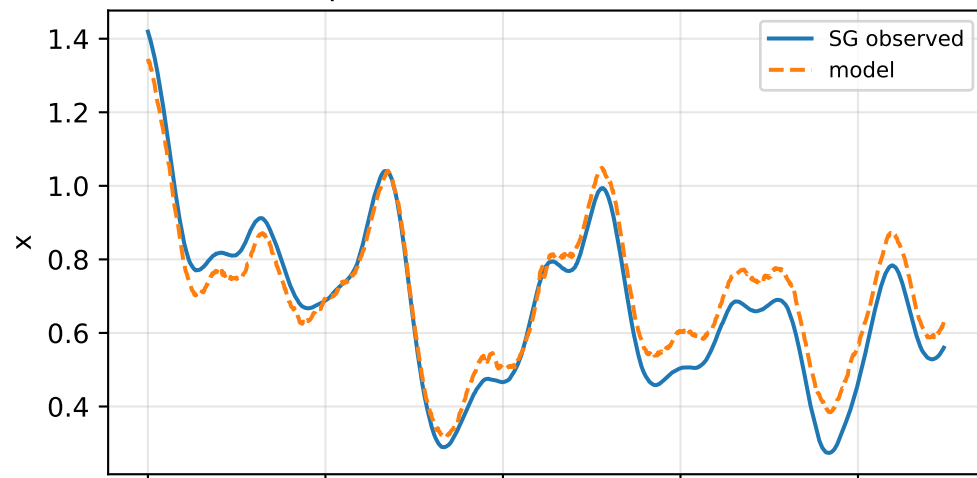


PSD excess model discrepancy correlation

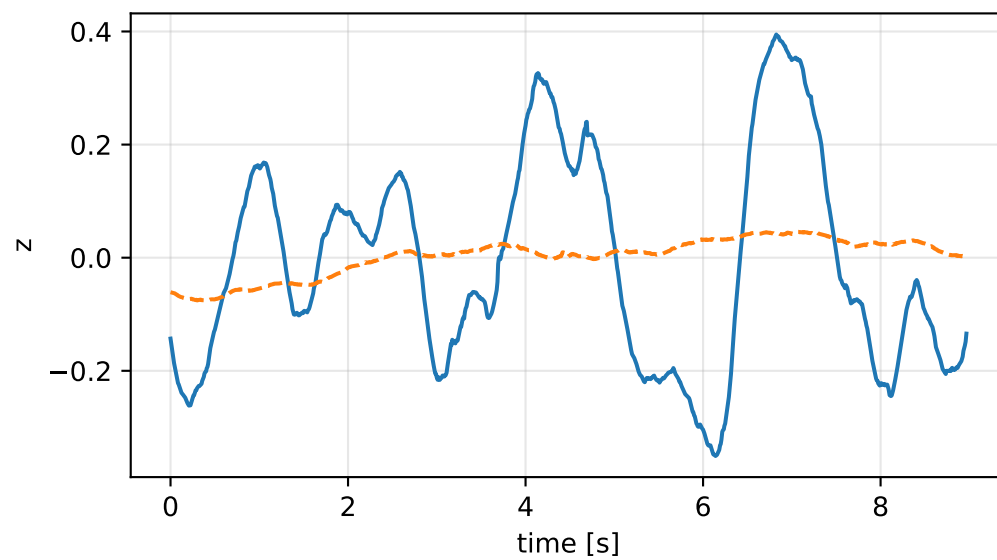
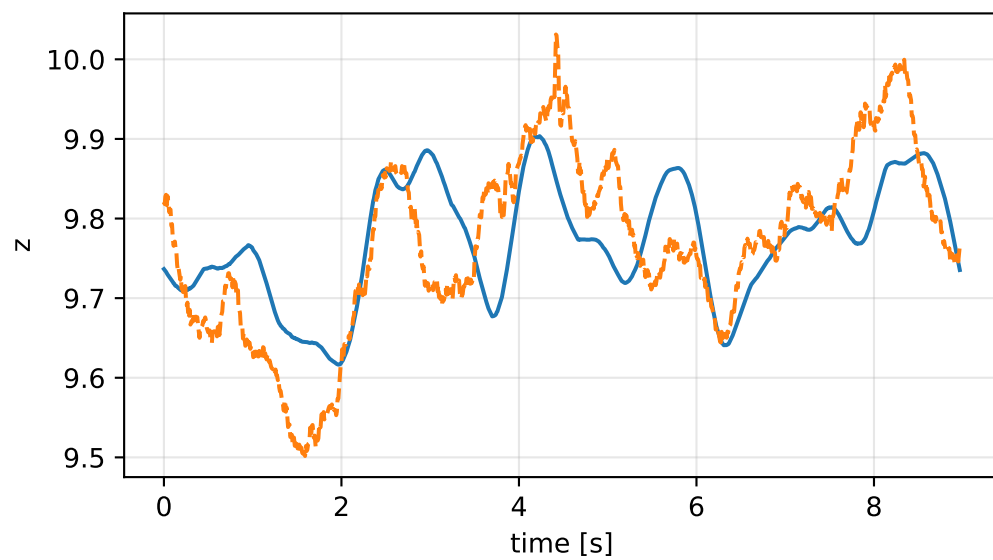
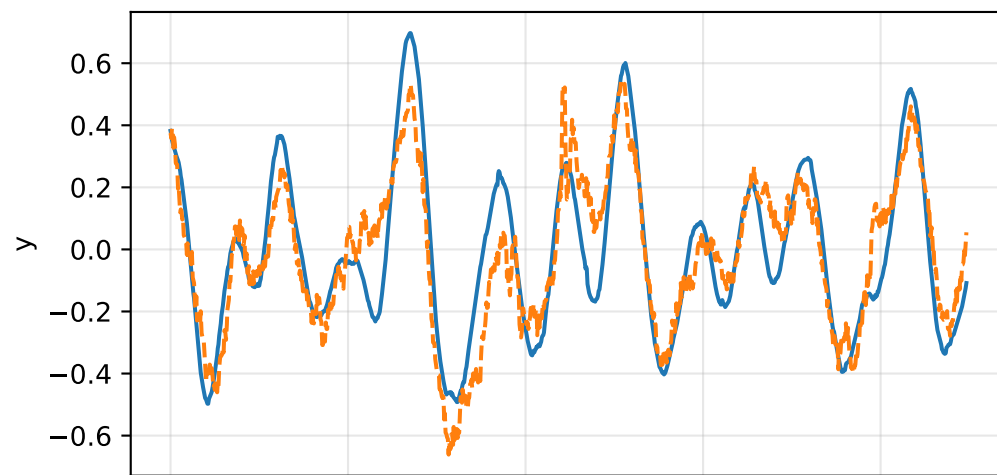
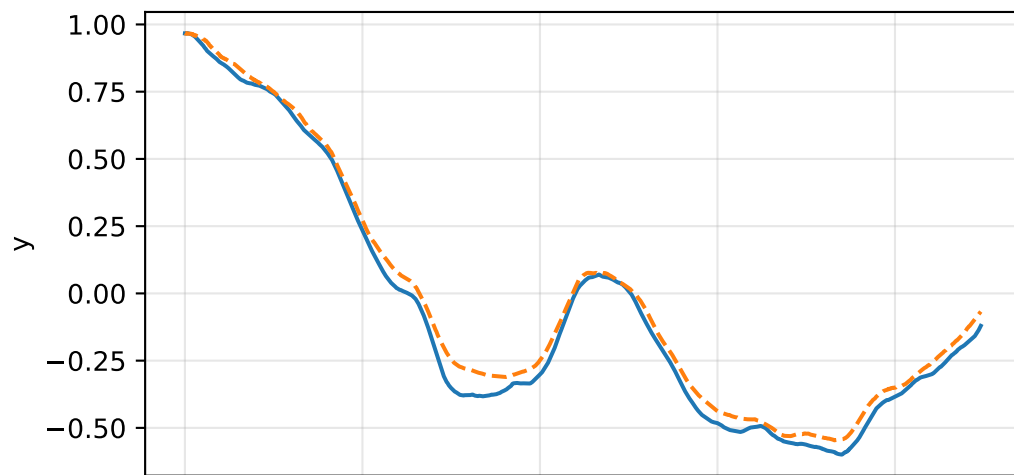
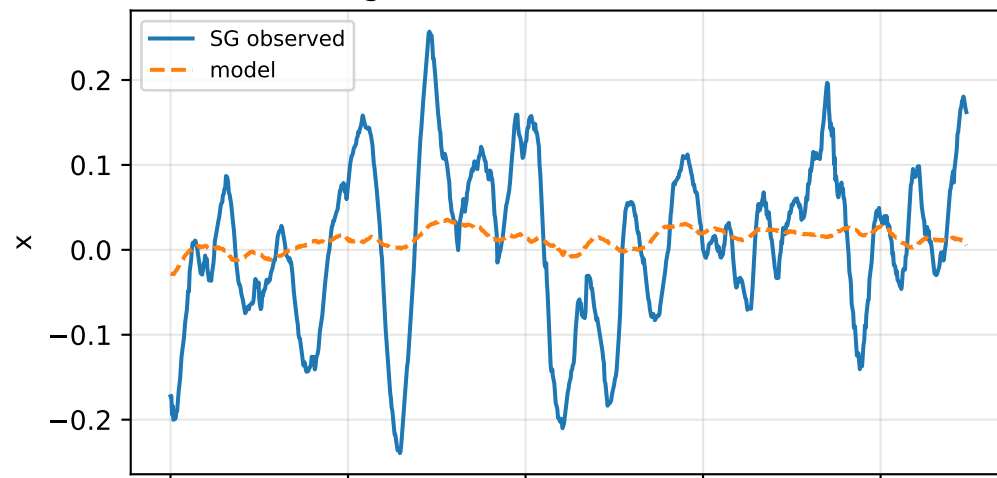


inertia_over_mass_yy_nominal: acceleration residual objective

specific acceleration [m/s^2]

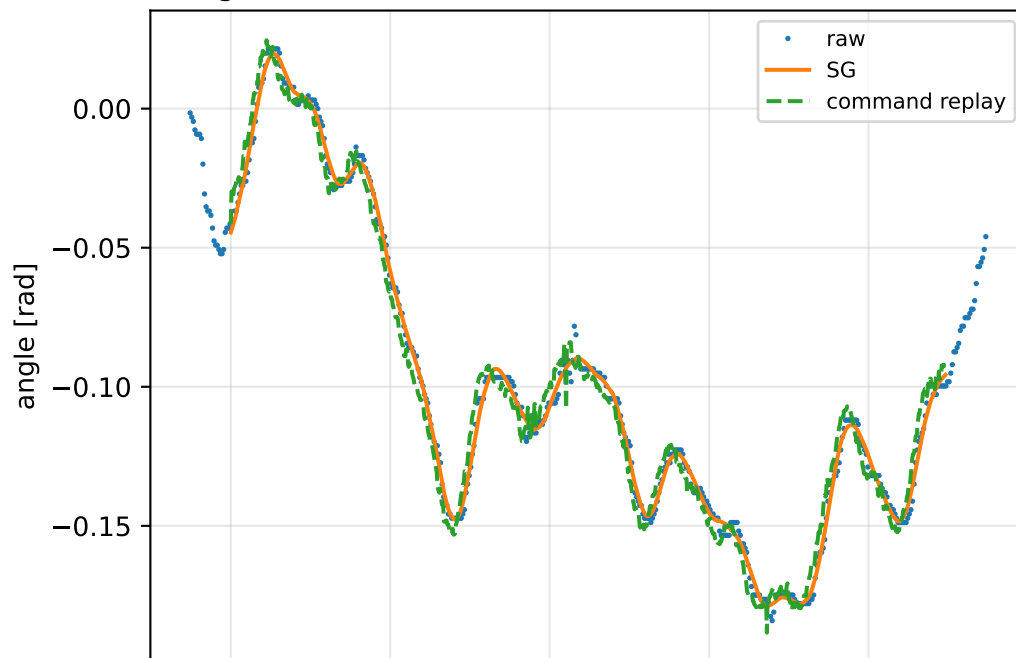


angular acceleration [rad/s^2]

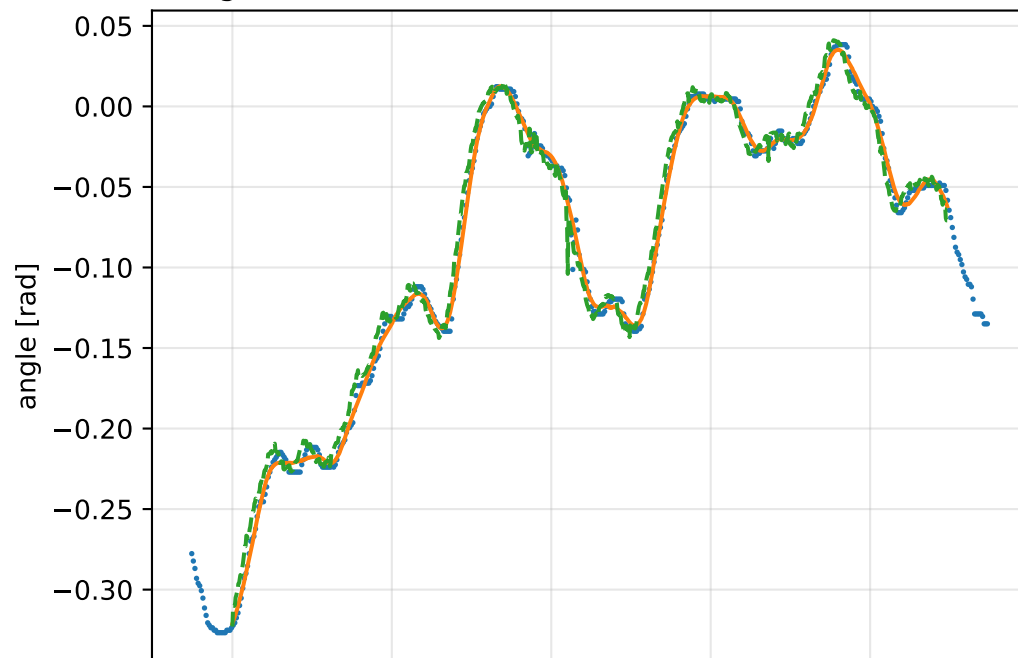


inertia_over_mass_yy_nominal: gimbal measurement audit (objective=measured_sg)

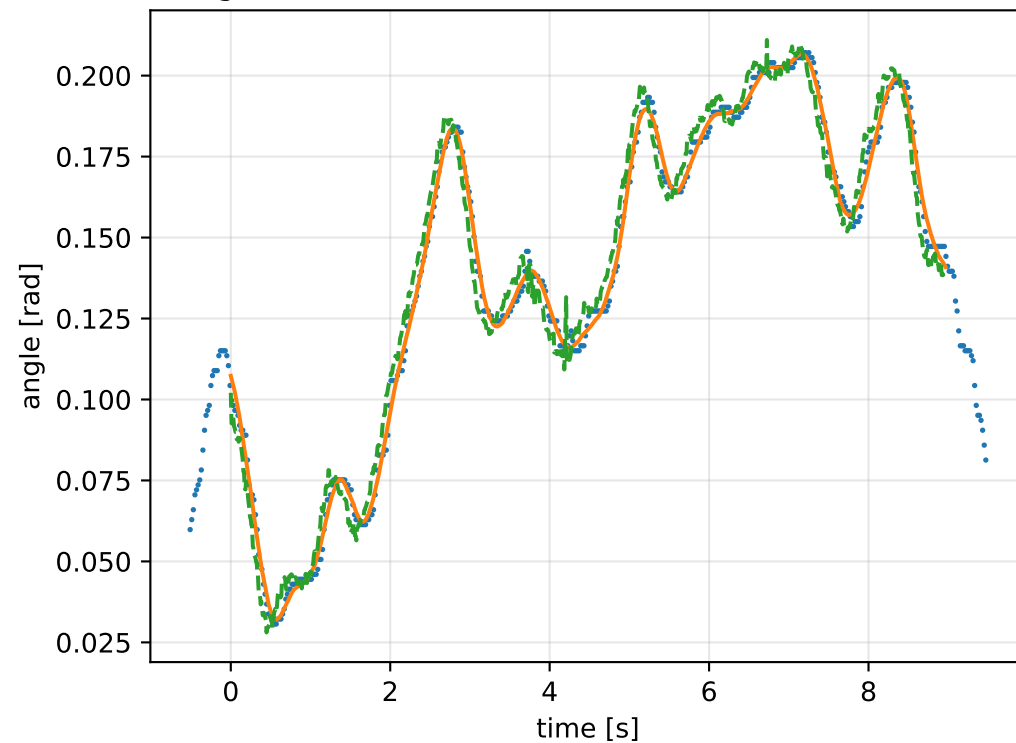
gimbal 1: RMSE=0.006345, max=0.01494 rad



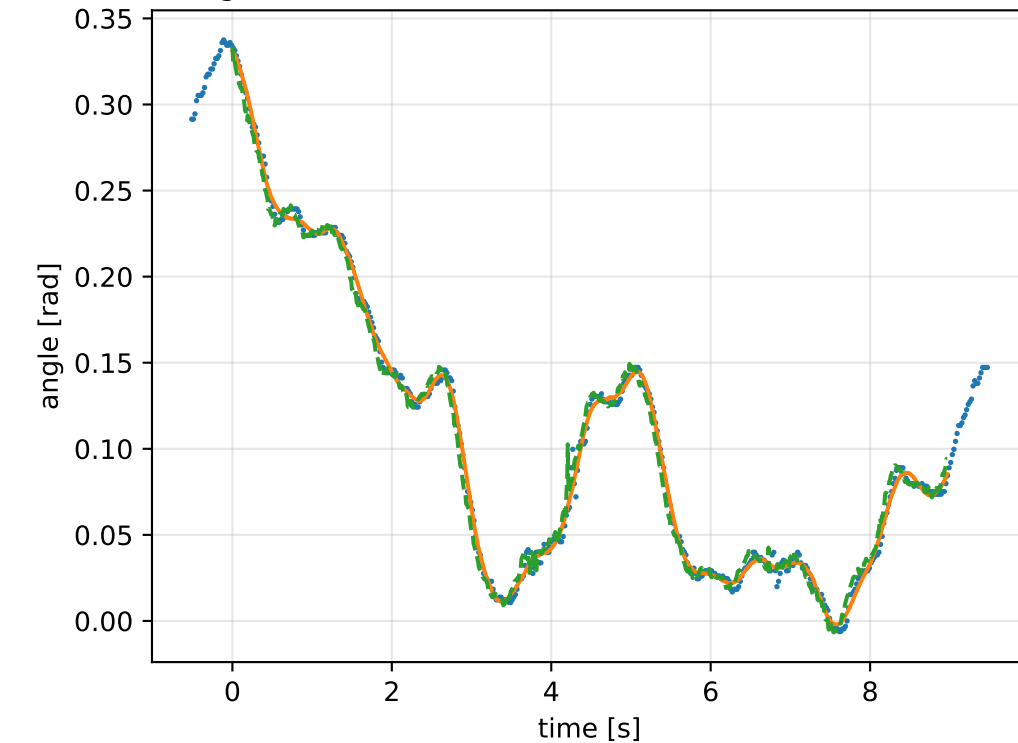
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

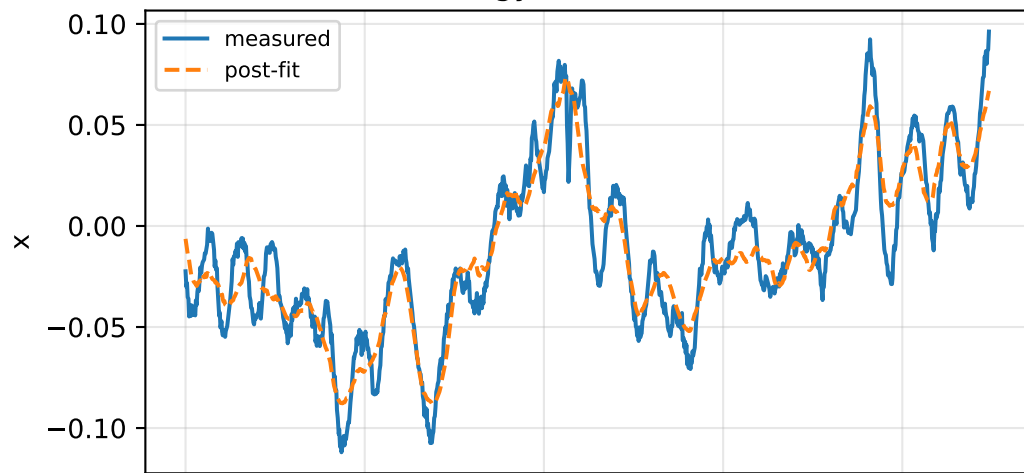


gimbal 4: RMSE=0.006321, max=0.03642 rad

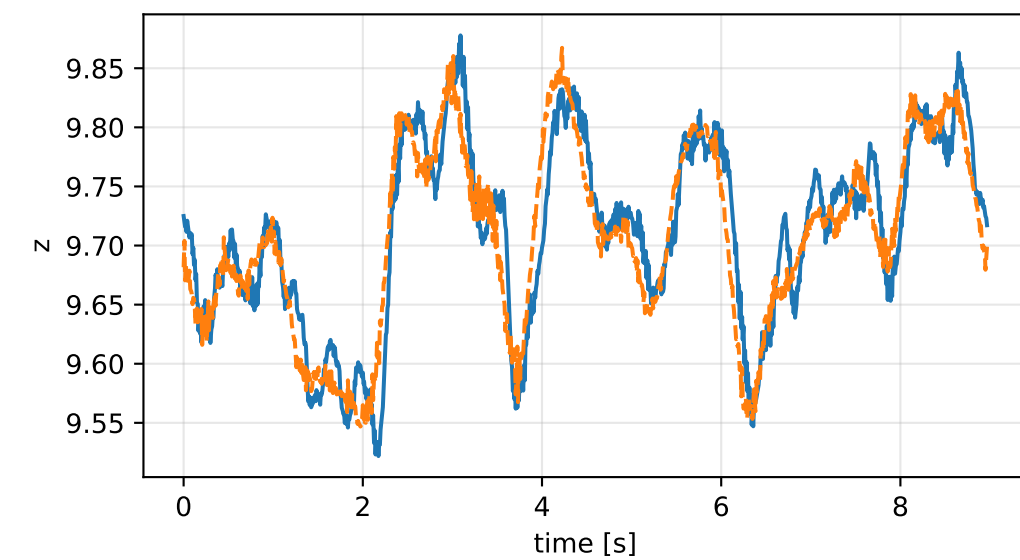
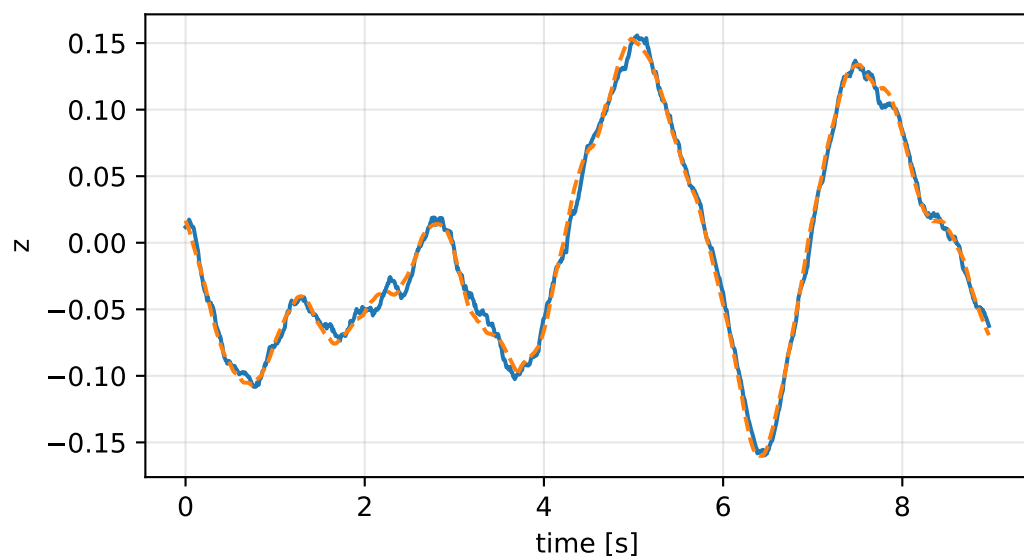
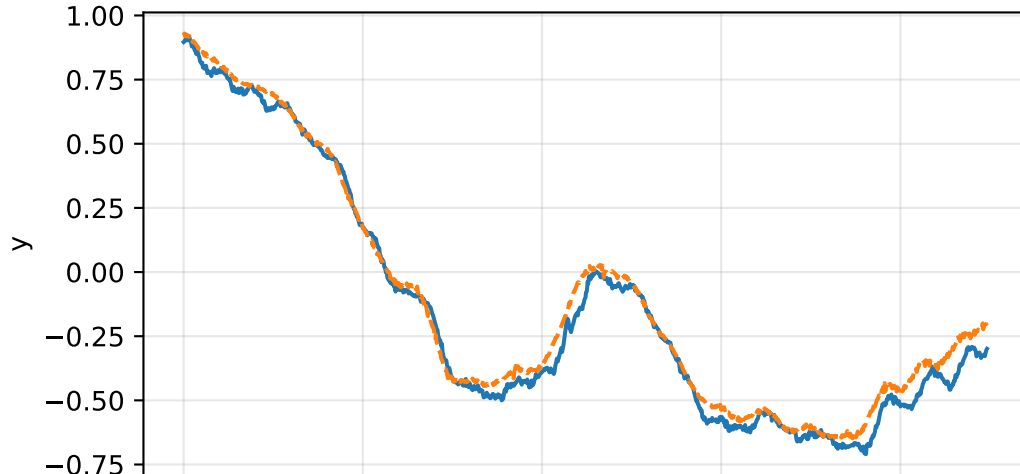
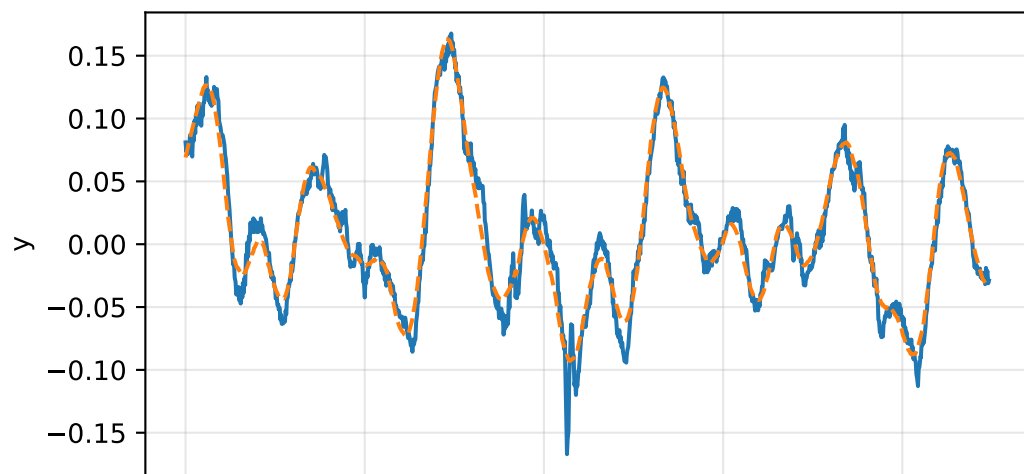
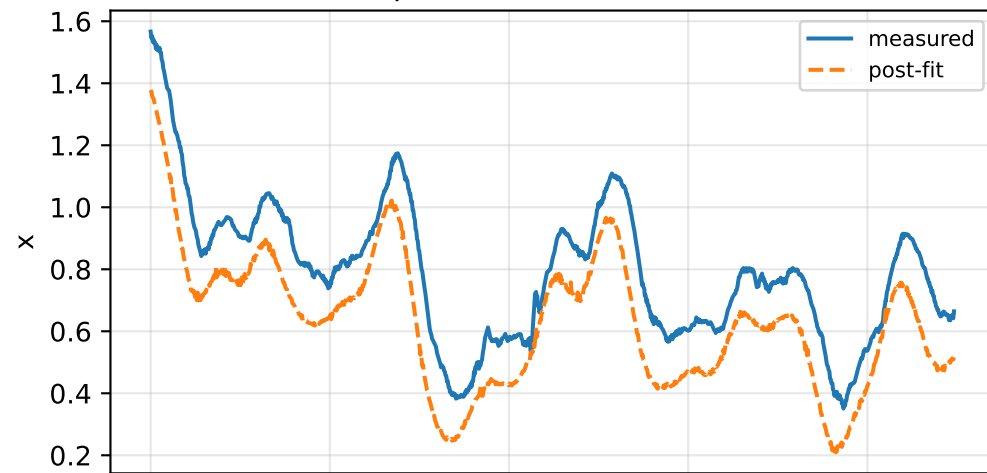


inertia_over_mass_yy_nominal: IMU comparison (diagnostic only)

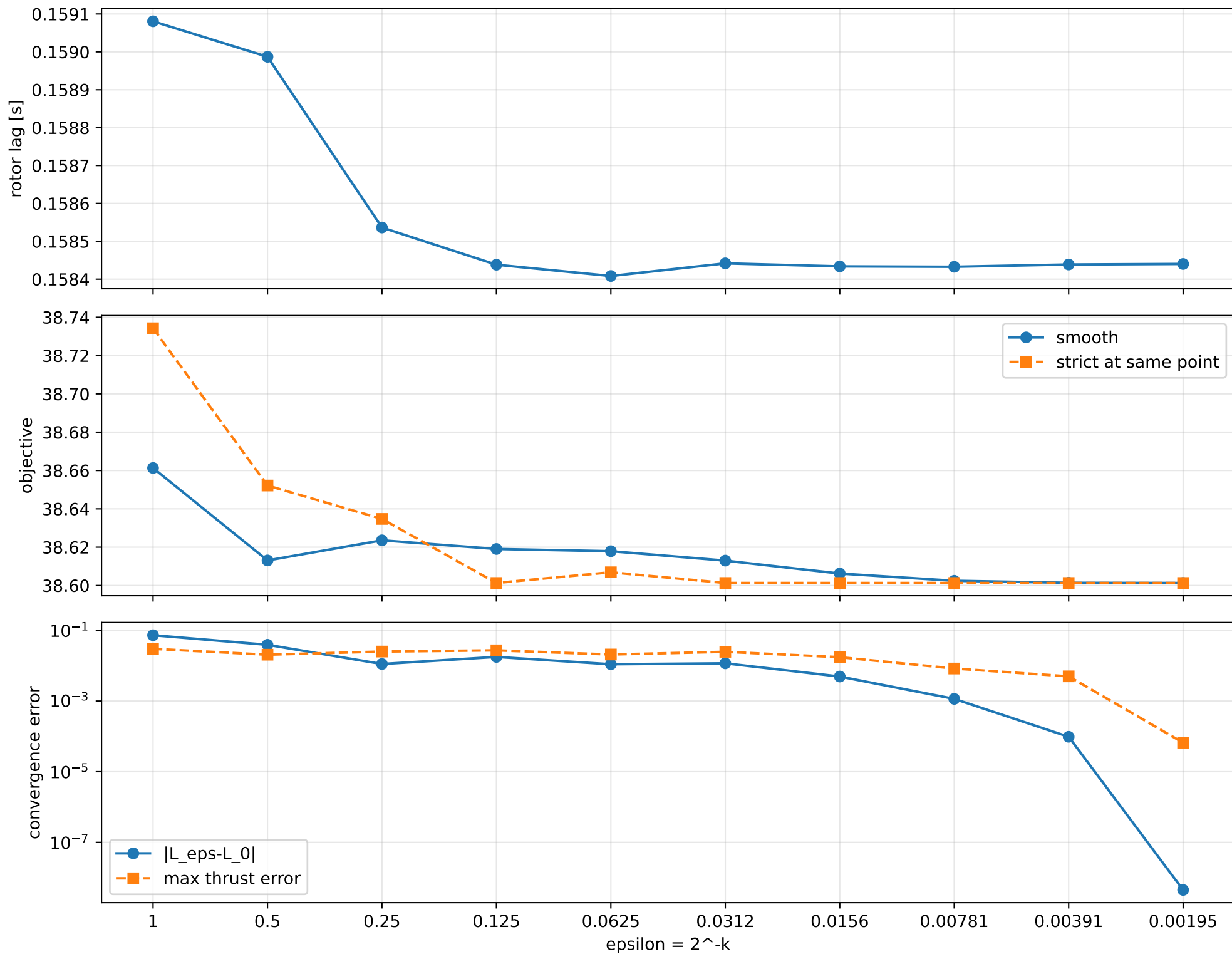
gyro [rad/s]



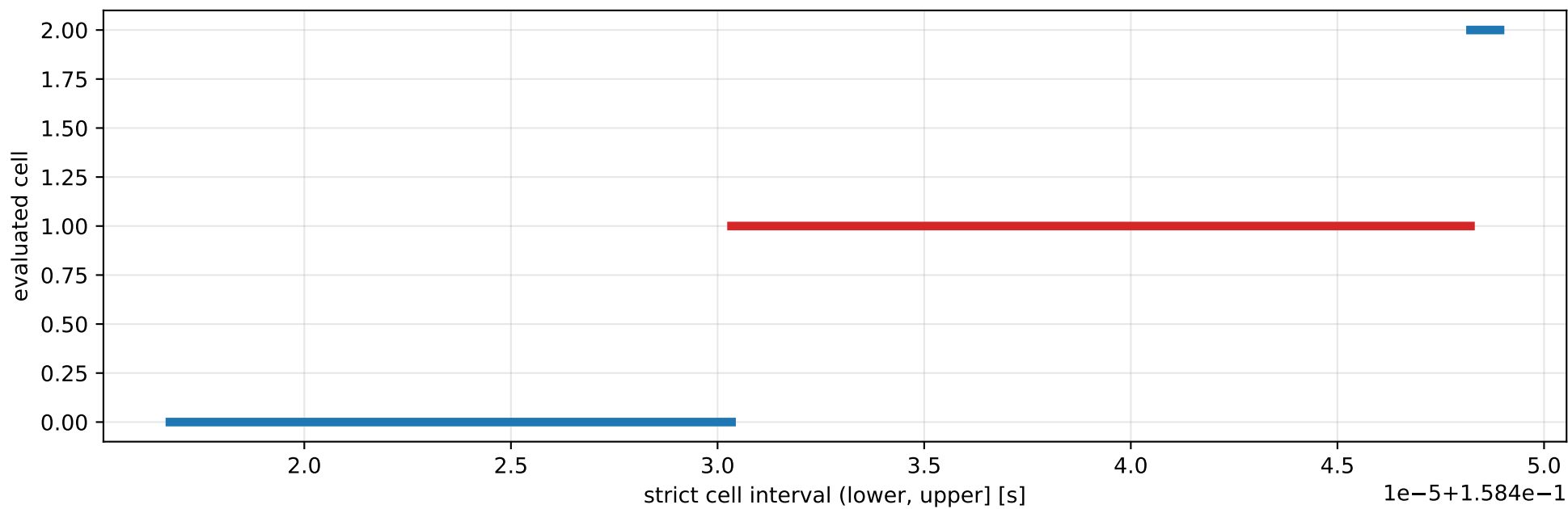
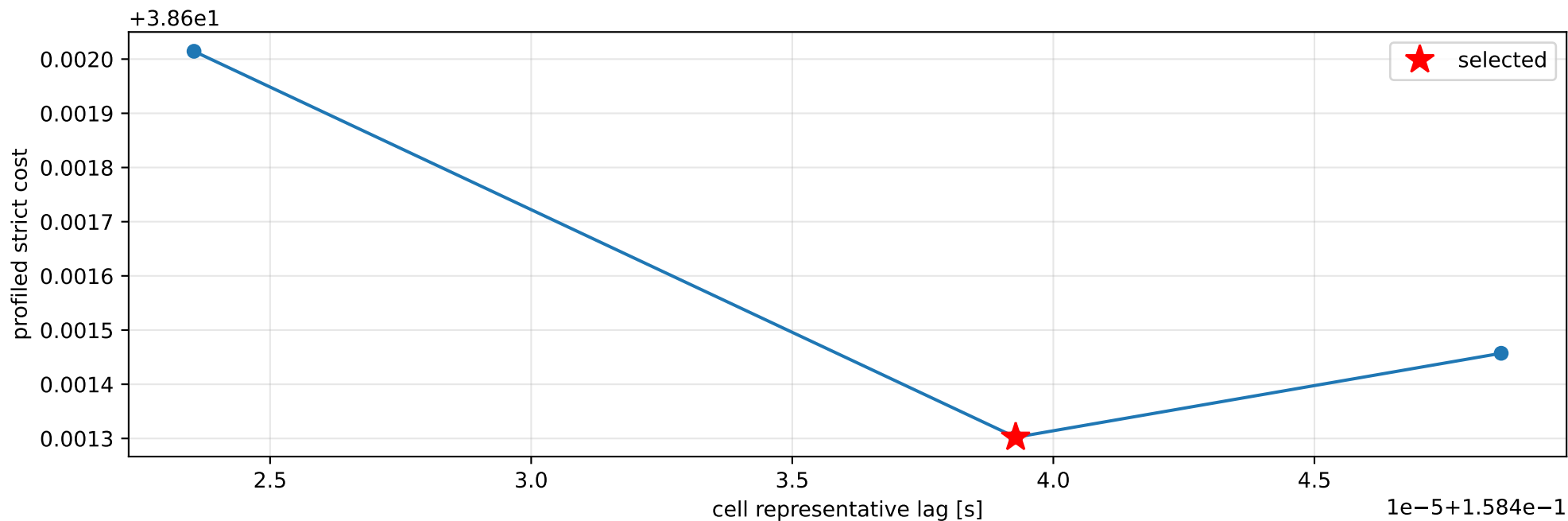
specific force [m/s^2]



inertia_over_mass_yy_nominal: smooth-to-strict continuation



inertia_over_mass_yy_nominal: exact strict-ZOH lag cells



selected (0.158430338, 0.158448219] s; final smooth 0.158440159 s; data boundary=False

inertia_over_mass_yy_nominal: parameter estimate

Case: inertia_over_mass_yy_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.2699582

inertia [kg m²]:

[[6.08201754 0.12221083 0.02781014]

[0.12221083 0.06269772 -0.13025045]

[0.02781014 -0.13025045 6.0563539]]

CoG body [m]: [-0.1994454 -0.02299295 -0.02998573]

force effectiveness: [1.53267524 0.14326047 0.0400485 1.29898205]

scale-free J/m [m²]:

[[2.67935222 0.05383836 0.01225139]

[0.05383836 0.02762065 -0.05738011]

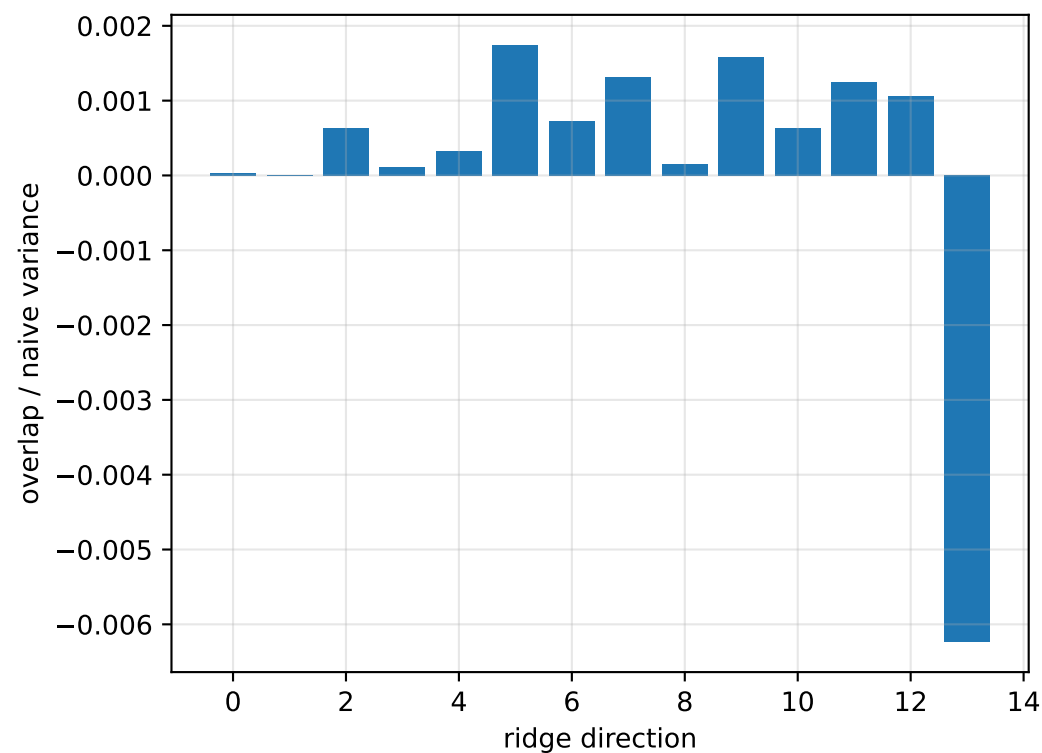
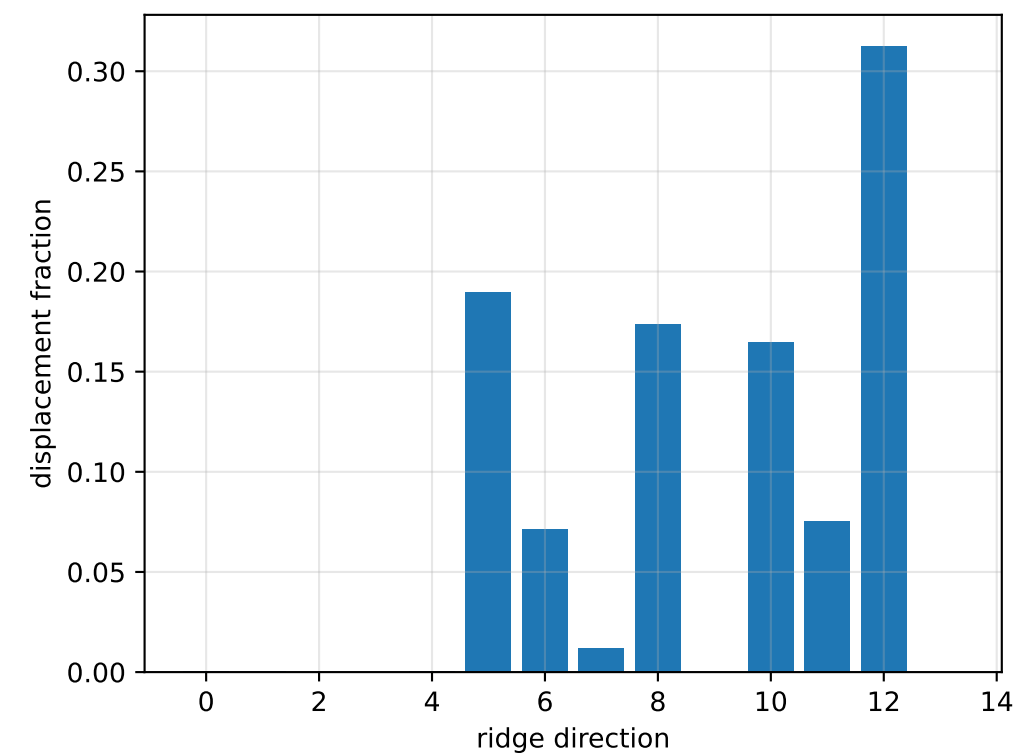
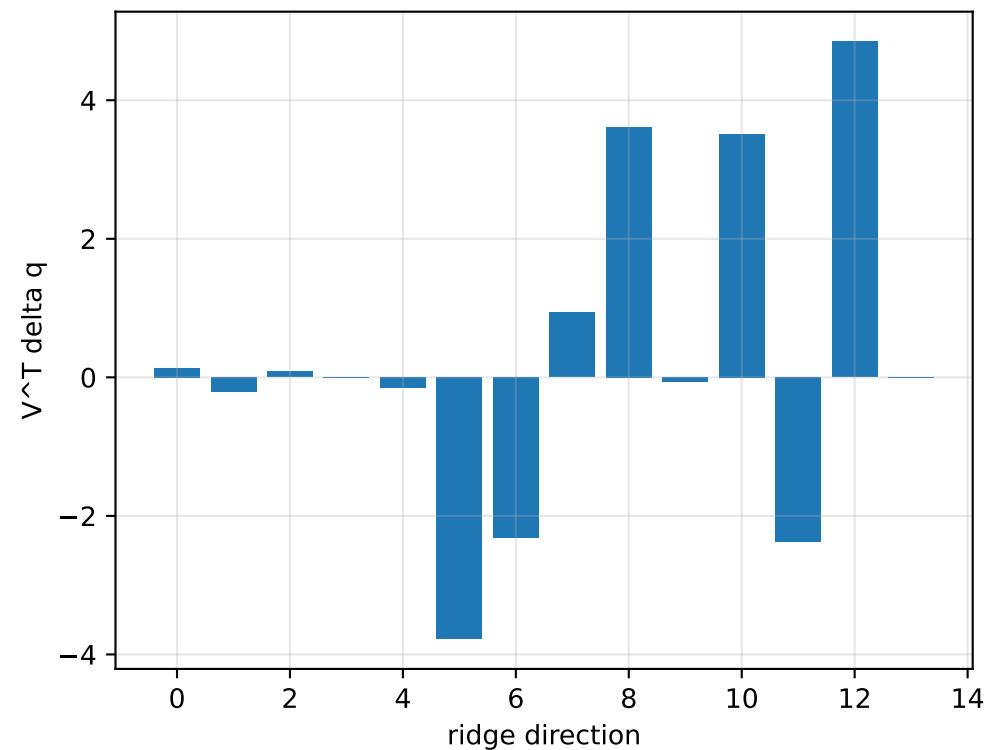
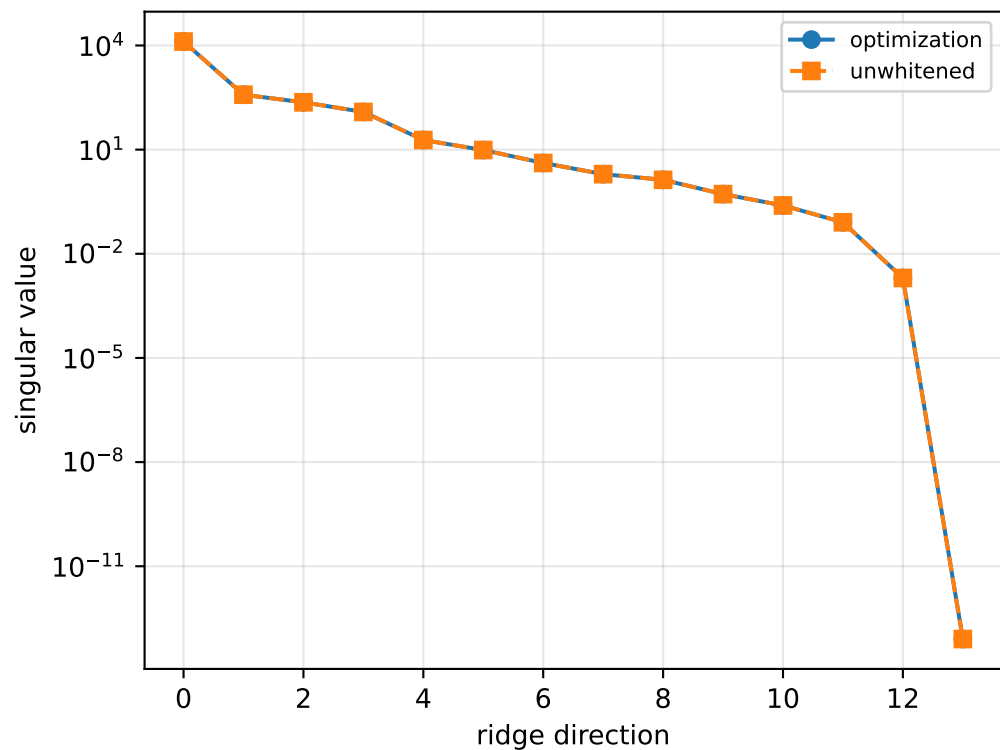
[0.01225139 -0.05738011 2.66804644]]

scale-free f/m: [0.67519976 0.0631115 0.01764284 0.57224932]

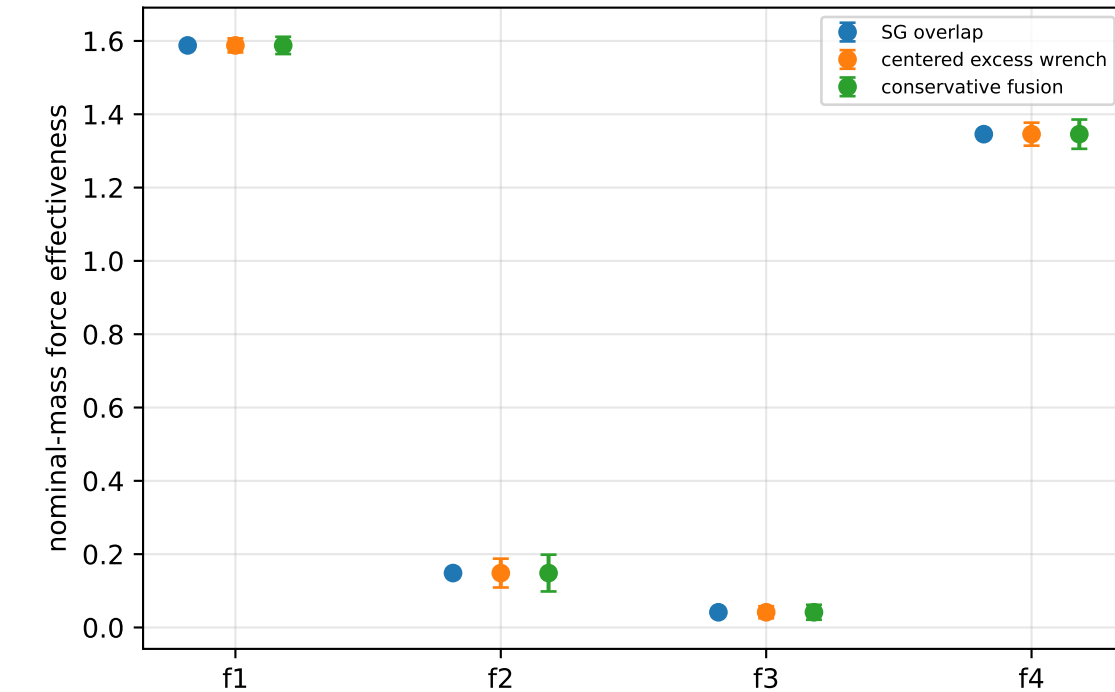
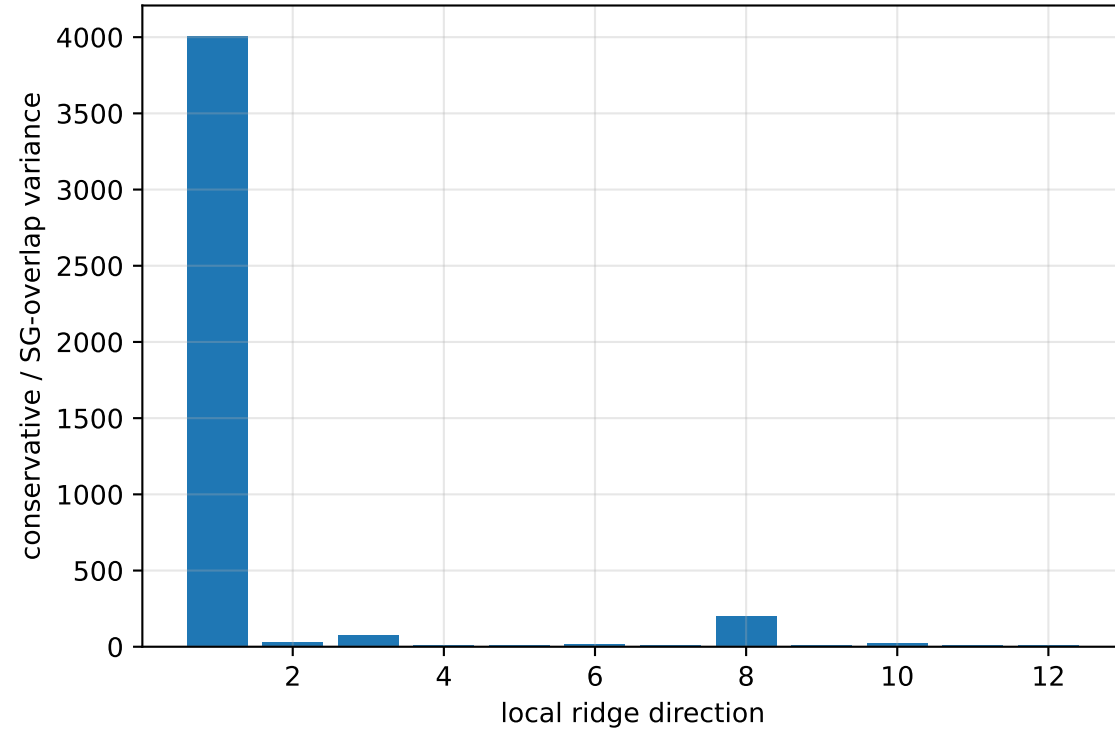
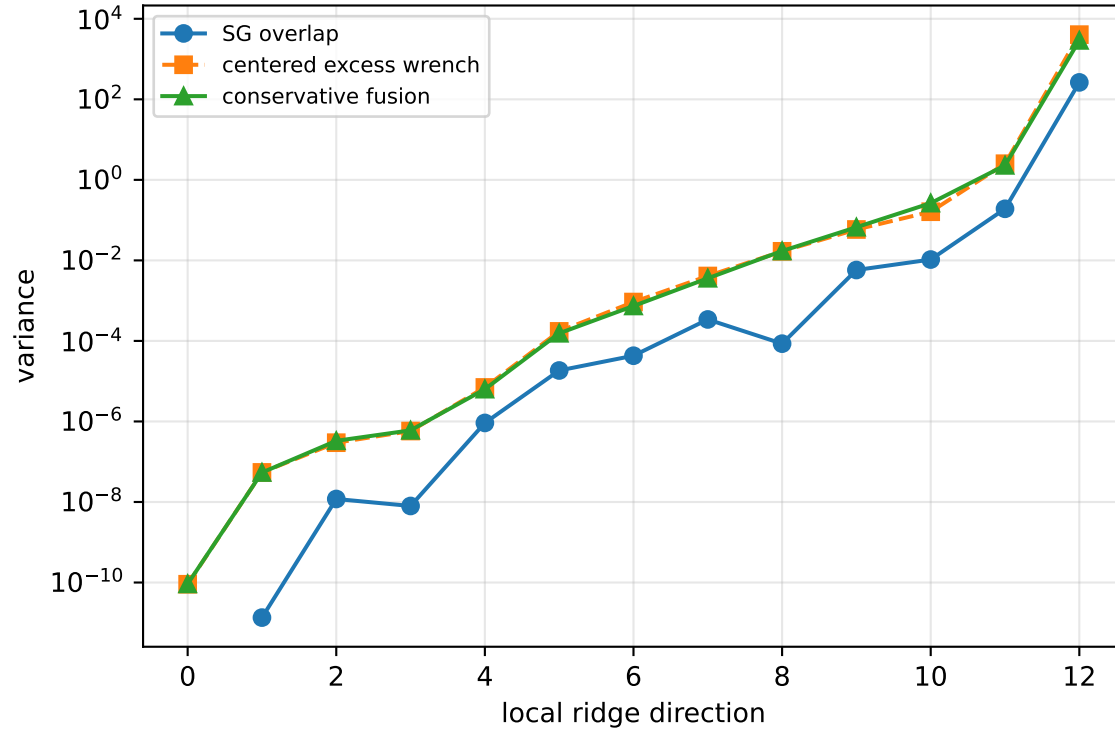
rotor lag cell: (0.158430338, 0.158448219) s

representative: 0.158439279 s

inertia_over_mass_yy_nominal: ridge spectrum (rank 13, nullity 1, || $\mathbf{v}$ ||=8.73e-14)



inertia_over_mass_yy_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 503.5$
wrench-to-acceleration recovery max error = $4.93\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

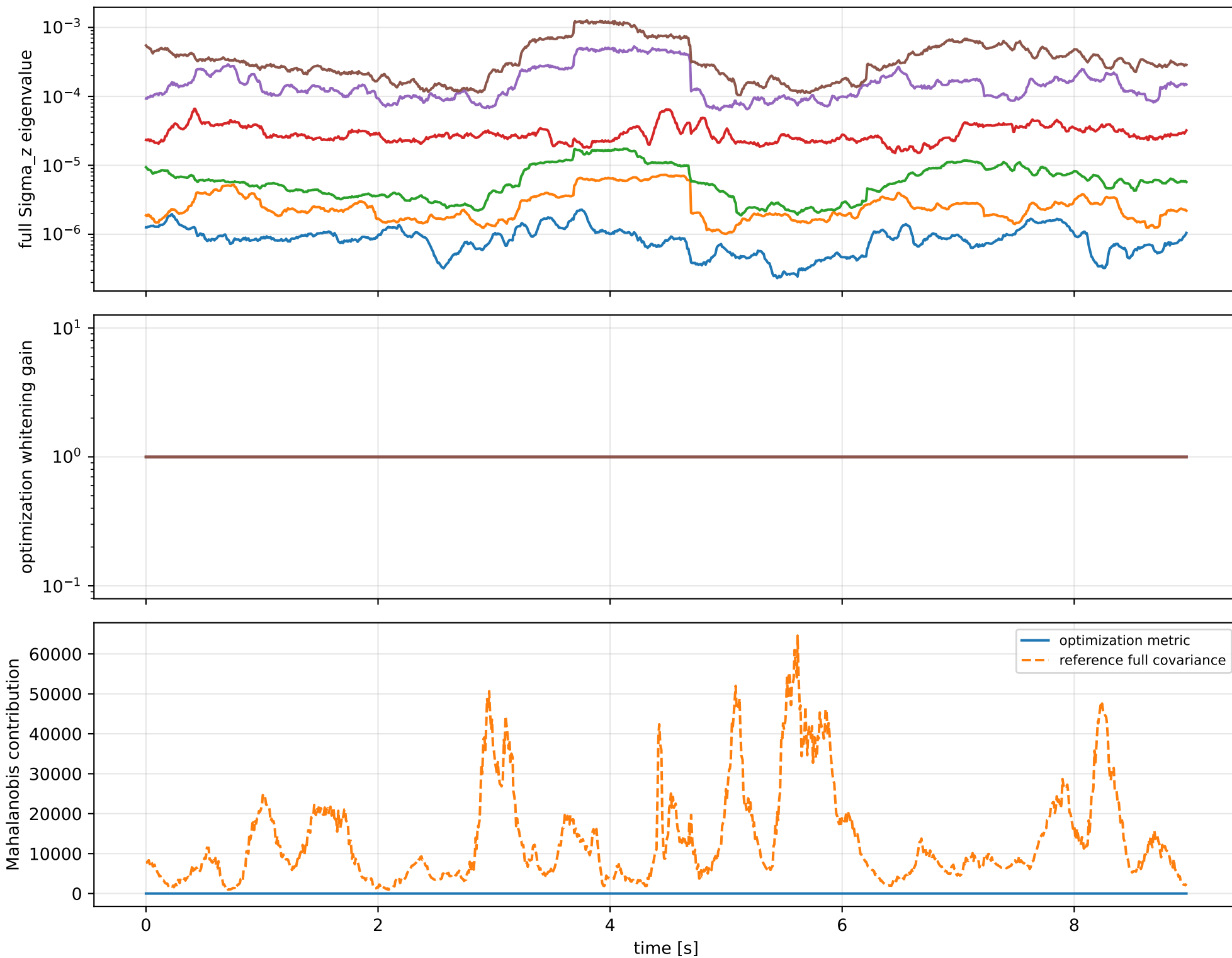
index 12: variance=2903, ratio=11.02
 $\log_{\text{mass}}=+0.132$; $\text{second_moment_diag_1}=-0.932$
 $\text{second_moment_diag_2}=+0.132$; $\text{second_moment_diag_3}=+0.141$
 $\text{second_moment_offdiag_12}=-0.0104$; $\text{second_moment_offdiag_13}=-0.0804$
 $\text{second_moment_offdiag_23}=-0.00294$; $\text{cog_x}=+3.89\text{e-}06$
 $\text{cog_y}=-1.2\text{e-}06$; $\text{cog_z}=+1.86\text{e-}06$
 $\log_{\text{force_eff_1}}=+0.132$; $\log_{\text{force_eff_2}}=+0.132$
 $\log_{\text{force_eff_3}}=+0.132$; $\log_{\text{force_eff_4}}=+0.132$

index 11: variance=2.287, ratio=11.97
 $\log_{\text{mass}}=+0.097$; $\text{second_moment_diag_1}=+0.0486$
 $\text{second_moment_diag_2}=+0.0828$; $\text{second_moment_diag_3}=-0.595$
 $\text{second_moment_offdiag_12}=+0.173$; $\text{second_moment_offdiag_13}=-0.741$
 $\text{second_moment_offdiag_23}=+0.123$; $\text{cog_x}=-0.000449$
 $\text{cog_y}=+0.000453$; $\text{cog_z}=+7.59\text{e-}06$
 $\log_{\text{force_eff_1}}=+0.097$; $\log_{\text{force_eff_2}}=+0.0731$
 $\log_{\text{force_eff_3}}=+0.0973$; $\log_{\text{force_eff_4}}=+0.0992$

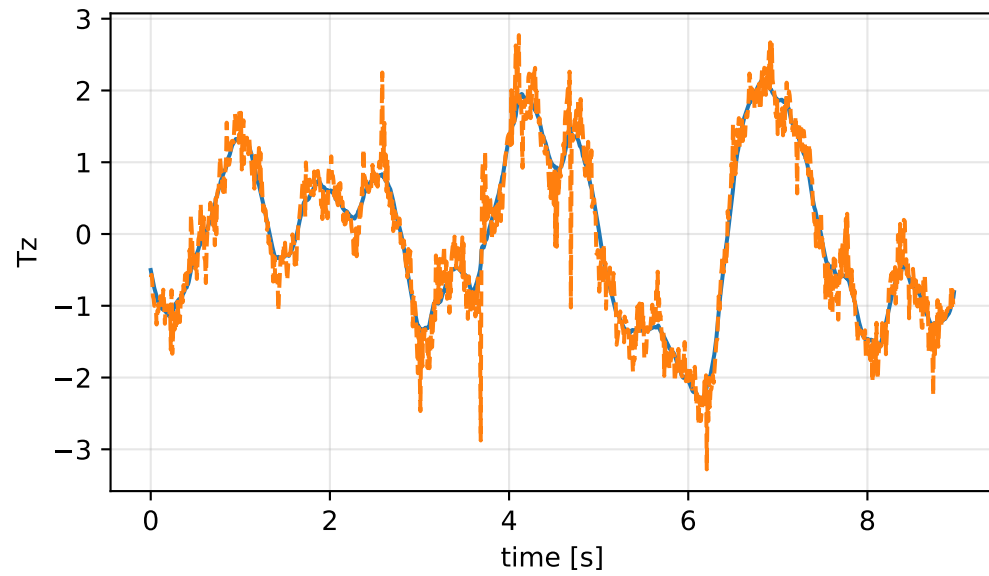
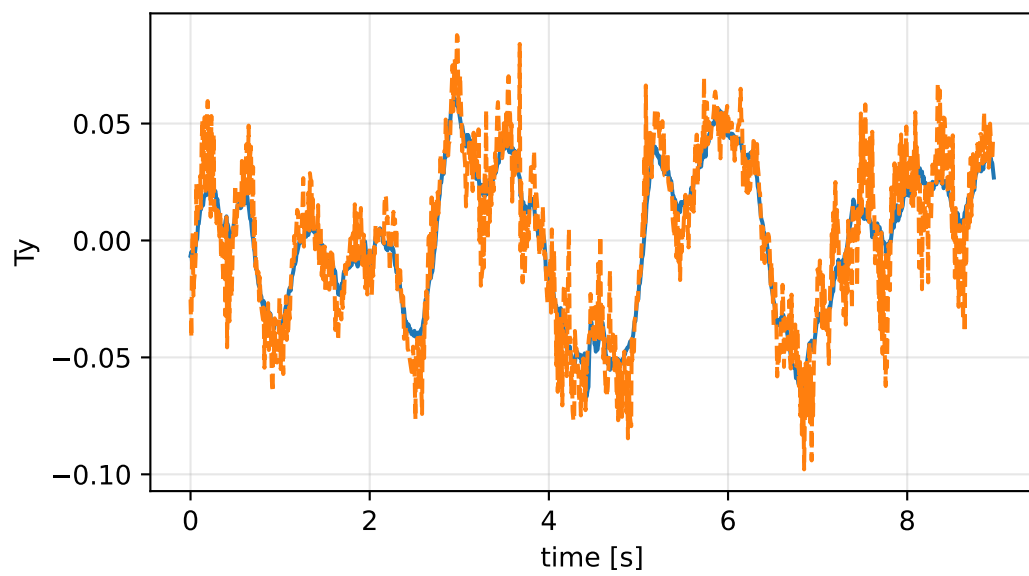
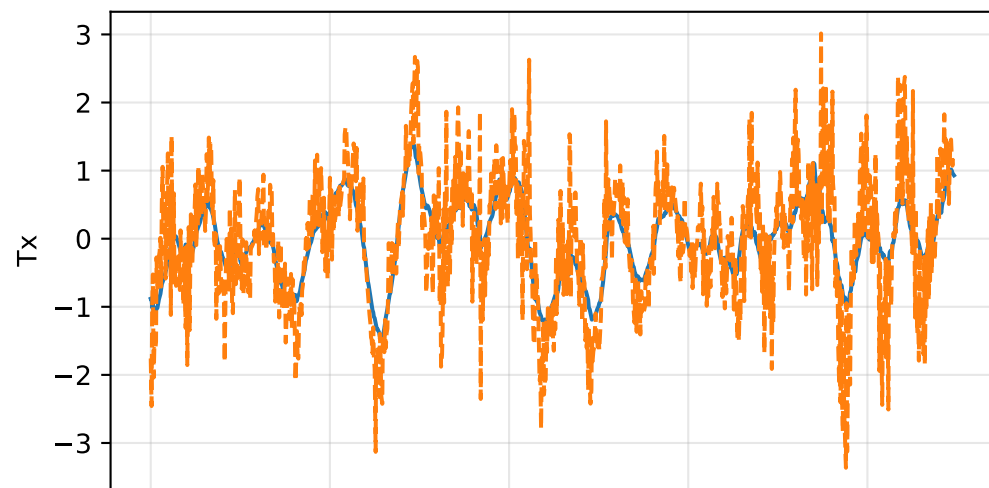
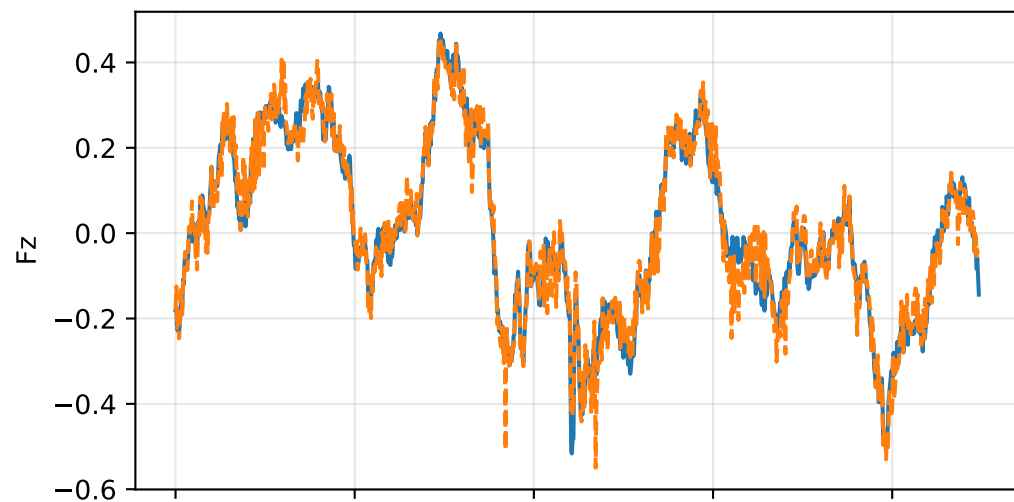
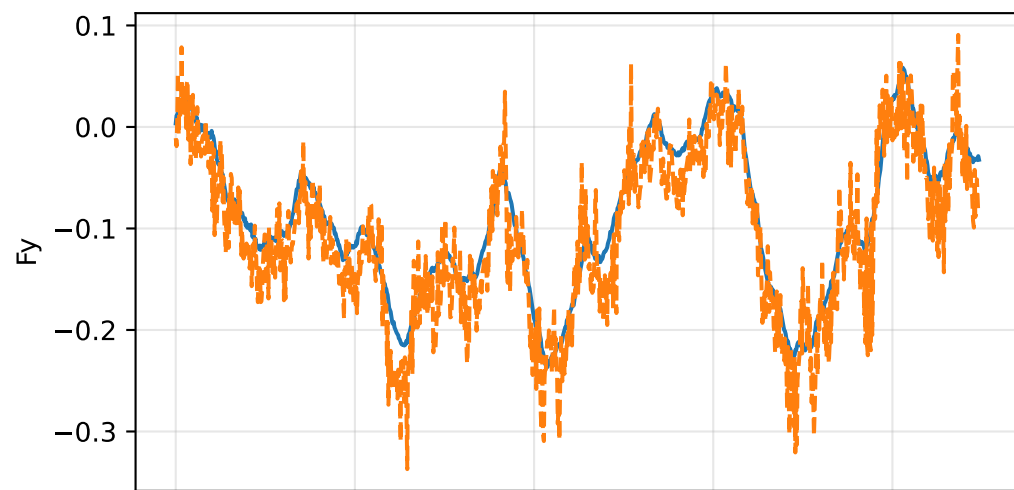
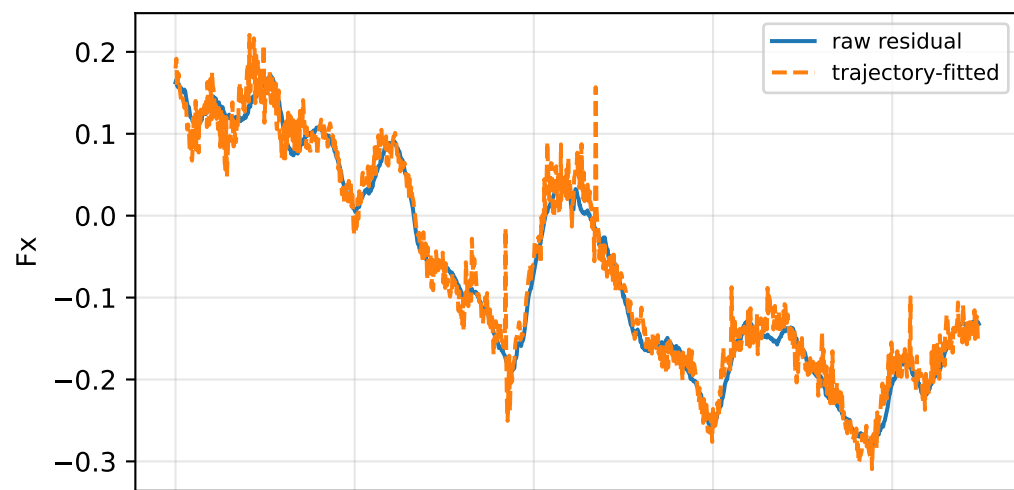
index 10: variance=0.2642, ratio=25.17
 $\log_{\text{mass}}=+0.238$; $\text{second_moment_diag_1}=-0.00835$
 $\text{second_moment_diag_2}=+0.333$; $\text{second_moment_diag_3}=-0.19$
 $\text{second_moment_offdiag_12}=-0.215$; $\text{second_moment_offdiag_13}=+0.116$
 $\text{second_moment_offdiag_23}=-0.0638$; $\text{cog_x}=-0.0126$
 $\text{cog_y}=+0.000811$; $\text{cog_z}=+0.00567$
 $\log_{\text{force_eff_1}}=+0.267$; $\log_{\text{force_eff_2}}=-0.162$
 $\log_{\text{force_eff_3}}=-0.749$; $\log_{\text{force_eff_4}}=+0.271$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_yy_nominal: optimization vs reference covariance



inertia_over_mass_yy_nominal: wrench / closure (force max $6.44\text{e-}15$, torque max $8.88\text{e-}16$)



inertia_over_mass_yy_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_yy_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 62834c577333da807f8b968827a18fec3976eb136afd16ab23bddf481b6a6c3e

data objective: 38.6013021727

prior objective: 1.32126260789e-09

total objective: 38.6013021741

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_yy_nominal (inertia_over_mass_m2)

components: yy

target: [0.02762065]

std: [1.e-06]

final: [0.02762065]

error: [5.14054979e-11]

standardized residual: [5.14054979e-05]

factor objective: 1.32126260789e-09